

Malware — Detailed Lecture Notes

JKSSB Computer Awareness

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42.1 What “malware” means

Malware is short for **malicious software**: any program written to harm a computer, steal from it, or misuse it. The official syllabus names **computer virus** and **latest anti-virus** explicitly, and the exam’s computer block tests the near neighbours as well: worm, Trojan, ransomware, spyware, adware, rootkit, keylogger, bot and botnet, backdoor, logic bomb, and cryptojacking.

Definition (Malware). *Any software designed to damage a computer, steal data from it, or misuse its resources without the owner’s consent.*

Term	Meaning in this lesson
Virus	Malware that attaches to a host file and needs human action to run.
Worm	Standalone, self-spreading malware that travels over networks.
Trojan (Trojan horse)	Malware disguised as useful software; does not replicate itself.
Ransomware	Malware that locks files and demands payment for release.
Spyware	Malware that secretly watches the user and steals data.
Adware	Malware that floods the machine with unwanted advertisements.
Keylogger	A spying tool that records every key pressed.
Rootkit	Malware that hides itself and other malware deep in the system.
Backdoor	A secret entry path into the system bypassing normal checks.
Bot / botnet	A remotely controlled machine / a network of such machines.
Logic bomb	Malicious code that waits for a trigger, then strikes.
Cryptojacking	Secret use of the victim’s computer to mine cryptocurrency.
Anti-virus	Software that detects, blocks, and removes known malware.

The rest of this lesson uses these terms in one consistent sense. A **virus** always needs a host file and a human launch. A **worm** needs neither. A **Trojan** pretends to be legitimate and does not replicate. **Ransomware** encrypts and demands money. **Spyware** hides and steals; **adware** shows ads. A **keylogger** records keystrokes. A **rootkit** conceals; a **backdoor** re-admits. A **botnet** is a controlled network (robot + network). A **logic bomb** waits for its trigger. **Cryptojacking** mines on the victim’s machine.

42.2 Virus, worm, and Trojan

A **virus** attaches itself to a host program or file and stays inert until a person opens, runs, or shares that file. Common carriers are email attachments, pen drives, and downloaded files. Once

launched it can damage, delete, or corrupt files. The defining limit is dependence: no host file and no human action, no spread.

Definition (Virus). *Malware that attaches to a host program or file and requires human action to execute and spread.*

A **worm** is standalone and self-replicating. It copies itself from machine to machine over network connections, exploiting security gaps, without any host file and without any user click. Its main harm is consumption: networks clog, systems slow, and services jam under the load of endless self-copies.

Definition (Worm). *Standalone malware that replicates and spreads across networks by itself, without a host file or human action.*

A **Trojan**, or Trojan horse, takes its name from deception: it arrives looking like something useful — a free game, a handy tool, a harmless document. The user installs or opens it willingly, and the hidden malicious part runs alongside. Unlike a virus or a worm, a Trojan does not replicate itself; each new machine needs its own act of trickery. Once inside, it can steal data, install spyware or ransomware, or open a backdoor for the attacker’s later return.

Definition (Trojan (Trojan horse)). *Malware disguised as legitimate software that tricks the user into installing it; it does not replicate itself.*

The three are best remembered as one table: a virus needs a host file and a user action; a worm needs neither; a Trojan needs the user’s trust and does not self-replicate. File damage points to a virus, network-wide slowdown with no clicks points to a worm, and a “useful” download that betrays the machine points to a Trojan.

Example 42.2.1. *An item describes a program that “spreads from computer to computer over the network without any user opening a file.” Trace the table: needs no host and no click means worm. If instead the stem says the malware “was installed as part of a free utility the user downloaded,” the deception plus no replication means Trojan.*

42.3 Ransomware

Ransomware locks the victim out of their own data, usually by encrypting files (or the whole screen), and then displays a demand: pay the ransom to receive the unlock key. The “pay to unlock” message, often with a countdown or a payment address, is its signature.

Ransomware usually arrives the Trojan way — a harmful attachment or link the user opens — and then encrypts documents, photos, and databases in minutes. The expected defences are regular **offline backups**, updated operating system and **anti-virus** software, and caution with attachments. The exam’s standing line is that paying the ransom does not guarantee recovery; the backup is what defeats the demand.

Definition (Ransomware). *Malware that encrypts the victim’s files (or locks the screen) and demands payment for the key to restore them.*

42.4 Spyware, adware, and the keylogger

Spyware is the silent watcher. It hides on the machine, observes browsing habits, logins, and personal details, and sends them to a third party. It deliberately causes no visible damage so the victim suspects nothing. It commonly arrives bundled with free downloads or deceptive installers, and its purpose is theft by observation: passwords, bank details, and identity data.

Adware is the noisy opposite. It floods the screen with unwanted advertisements — pop-ups, banners, redirected searches — to earn money from ad traffic. Heavy adware also slows the machine and may carry spyware along with it, so the exam treats persistent pop-ups as more than a nuisance.

A **keylogger** is a focused spying tool: it records every key pressed and forwards the log to the attacker, harvesting usernames, passwords, and card numbers. Keyloggers exist as software and as small hardware devices fitted between the keyboard and the computer.

Definition (Spyware vs adware). *Spyware hides and steals information quietly; adware displays unwanted advertisements openly. A keylogger is a spying tool that records keystrokes.*

Example 42.4.1. *An item reports “constant pop-up advertisements and redirected searches.” The visible ads mean adware. If instead the item reports “passwords stolen although nothing on screen looked wrong,” the silent theft means spyware, and if it specifies “every keystroke was recorded,” the tool is a keylogger.*

42.5 Rootkit and backdoor

A **rootkit** is malware whose job is concealment. It lodges deep in the system, often with administrator-level control, and hides files, processes, and network activity from the user and from ordinary detection tools — including hiding other malware.

A **backdoor** is the secret entry a rootkit or Trojan protects: a path into the system that bypasses normal login and security checks, left open so the attacker can return at will. The pairing to remember is that Trojans and rootkits commonly install backdoors; the rootkit hides the intruder, and the backdoor keeps the door open.

Definition (Rootkit and backdoor). *A rootkit conceals an intruder and its malware deep in the system; a backdoor is a hidden entry path that bypasses normal authentication.*

42.6 Bot and botnet, logic bomb, cryptojacking

A **bot** is an ordinary computer that has been compromised and is now obeyed remotely; a **botnet** is a network of such machines (the word joins robot + network) under one master. Botnets send spam, spread further malware, and launch large-scale attacks that no single machine could mount. The owner typically notices only slowness.

A **logic bomb** is malicious code already planted that waits for a trigger — a date, an event, or a specific action — and strikes only when the condition is met. It does not spread; it waits.

Cryptojacking is theft of computing power: the attacker secretly runs cryptocurrency mining on the victim’s machine, so the computer runs hot and slow while the attacker collects the earnings. Like a bot, its outward sign is an otherwise unexplained slowdown.

42.7 How malware reaches a machine, and how to stop it

Malware reaches users through a short list of routes: harmful email attachments and links, infected pen drives, bundled free-software installers, downloads from untrusted sites, and unpatched system software with known gaps.

The defences form a matching list. **Anti-virus** software detects, blocks, quarantines, and removes known malware; both it and the operating system must be kept updated, since each update closes gaps and refreshes the list of known threats. A **firewall** filters incoming and outgoing

network traffic and blocks suspicious connections, but it is a barrier, not a cleaner: it does not remove malware already inside. Safe habits complete the set: do not open unknown attachments or links, scan pen drives before use, download only from trusted sources, and keep regular **offline backups** so ransomware loses its leverage.

42.8 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. A virus needs a host file **and** a human action; a worm needs **neither**; a Trojan disguises itself and **does not replicate**.
2. Spyware **hides and steals**; adware **shows ads**. A keylogger records **keystrokes**.
3. Ransomware **encrypts and demands payment**; its signature is the “pay to unlock” message, and the defence is the **offline backup**.
4. A rootkit **hides** the intruder; a backdoor is the **secret re-entry**. Trojans commonly install backdoors.
5. A botnet is a **network** of remotely controlled machines (robot + network), used for spam and large attacks.
6. A logic bomb **waits for a trigger**; cryptojacking **mines cryptocurrency** on the victim’s machine.
7. Anti-virus **removes** malware; a firewall **filters** traffic and does not clean an infected machine.

Example 42.8.1. *An item asks which malware “encrypts the user’s files and demands money for the decryption key.” The lock plus the demand means ransomware — not a worm (self-spreading), not adware (ads), not spyware (silent theft). If the stem instead says the program “waits until a fixed date and then deletes files,” the waiting trigger means a logic bomb.*

Exercises

1. Distinguish a virus from a worm from a Trojan on three points: host file, self-spreading, and user action.
2. Contrast spyware with adware. Where does a keylogger fit?
3. Explain what ransomware does and name the defence that defeats its demand.
4. Distinguish a rootkit from a backdoor, and explain how they work together.
5. Define bot/botnet, logic bomb, and cryptojacking in one sentence each, naming each type’s outward sign.

Solutions

1. A virus needs a host file and a human launch and does not spread by itself; a worm is standalone and self-spreading over networks with no host and no click; a Trojan pretends to be legitimate software, is installed willingly, and does not replicate itself.
2. Spyware hides and secretly steals user data; adware openly floods the machine with unwanted ads. A keylogger is a focused spying tool that records every keystroke to steal credentials.
3. Ransomware encrypts the victim's files (or locks the screen) and demands payment for the unlock key. Regular offline backups defeat the demand, since the data can be restored without paying.
4. A rootkit hides the intruder and its malware deep in the system; a backdoor is a secret entry bypassing normal checks. Trojans and rootkits install backdoors so the attacker can return unseen.
5. A bot is a remotely controlled compromised machine and a botnet is a network of them (sign: slowness during spam or attacks); a logic bomb waits for a trigger such as a date and then strikes (sign: a timed surprise); cryptojacking secretly mines cryptocurrency on the victim's machine (sign: hot, slow computer).